

Structure, functional annotation and inhibition of a putative D-alanine-D-alanine ligase protein

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1 Protein structure prediction

Homology models based on eight templates were initially built for comparison (Table 1), candidate templates being chosen based on sequence similarity to query and to each other, oligo-state, and ligand-binding [1, 2]. The 7u9k.1.A template, PDB entry 7U9K *Staphylococcus aureus* scores D-alanine-D-alanine ligase (DDL) in complex with ATP, D-ala-D-ala, Mg^{2+} and K^+ , was ultimately selected, having been found to produce the model with the highest GMQE and QMEANDisCo Global scores together (Table 1) as well as better QMEANDisCo Local and QMEAN Z-Scores, compared to other models (Figure 1) [3, 4].

Table 1. Template information and associated model global quality metrics.

Template					Model	
Template	Identity (%)	Experimental method	Oligo state	Ligands	GMQE	QMEANDisCo Global
3i12.1.A D-alanine-D-alanine ligase A	31.54	X-ray, 2.2Å	homo-dimer	2 x ADP	0.75	0.72 ± 0.05
7u9k.1.A D-alanine--D-alanine ligase	36.45	X-ray, 2.0Å	homo-dimer	1 x DAL-DAL, 4 x K, 2 x MG, 1 x ADP, 1 x ATP	0.75	0.74 ± 0.05
2i80.1.A D-alanine-D-alanine ligase	36.58	X-ray, 2.2Å	homo-dimer	2 x G1L	0.73	0.72 ± 0.05
2zdg.1.B D-alanine--D-alanine ligase	36.56	X-ray, 2.2Å	homo-dimer	2 X ADP	0.70	0.71 ± 0.05
8evx.1.A D-alanine--D-alanine ligase A	32.41	X-ray, 1.5Å	homo-dimer	2 x ADP, 2 x DS0, 4 x MG, 2 x K	0.71	0.72 ± 0.05
3v4z.1.B D-alanine--D-alanine ligase	31.79	X-ray, 2.7Å	homo-dimer	None	0.67	0.70 ± 0.05
1e4e.1.A VANCOMYCIN/TEICOPLANIN A-TYPE RESISTANCE PROTEIN VANA <i>D-alanyl-D-lacate ligase</i>	30.18	X-ray, 2.5Å	hetero-tetramer	4 x ADP, 4 x PHY, 8 x MG	0.70	0.71 ± 0.05
3r5x.1.B D-alanine--D-alanine ligase	32.75	X-ray, 2.0Å	homo-dimer	3 x CA, 2 x ATP	0.65	0.66 ± 0.05

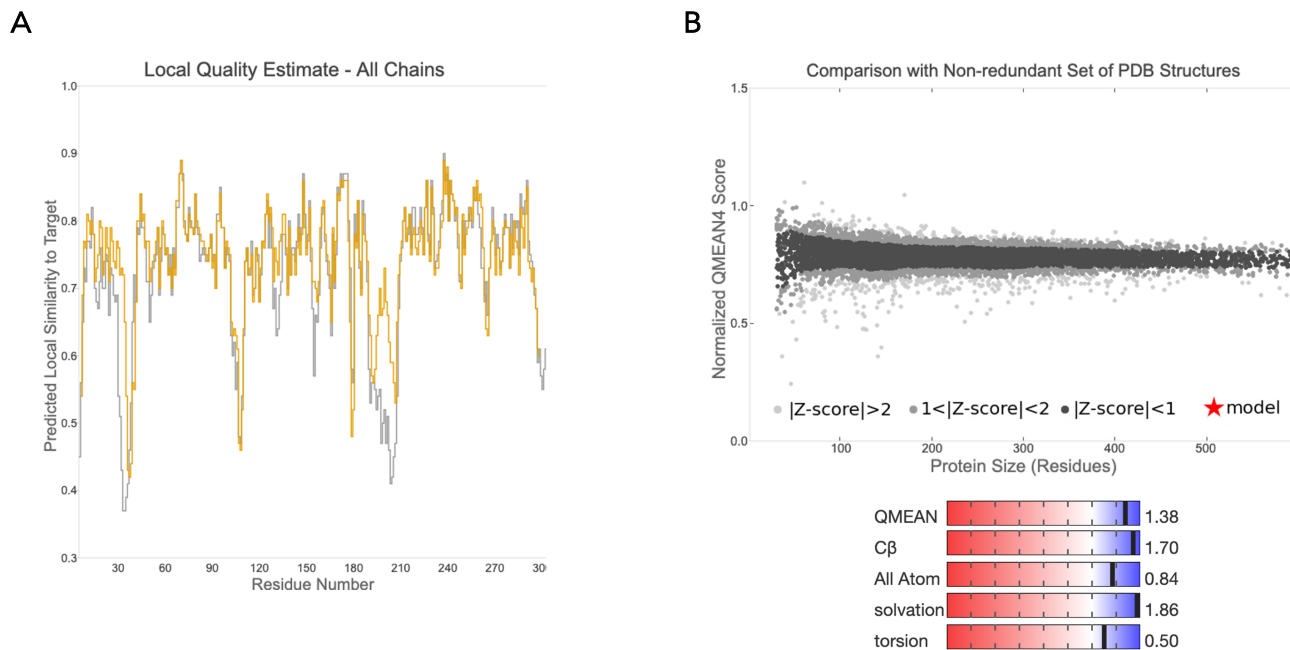


Figure 1. Local quality scores for the 7u9k.1.A-derived model. QMEANDisCo Local (A) and QMEAN Z-Scores (B).

An AI-predicted protein structure was generated using AlphaFold3 due to inclusion of ions and ability to produce All-atom predictions. *S. Aureus* DDI was used as reference for co-factor inclusion and while 7U9K has 1 Mg²⁺ and 2 K⁺ cations present in the active site, the literature indicates that 2 Mg²⁺ and only 1 K⁺ are generally present, including Becker *et al.* (2023) which describes the second Mg²⁺ as having been replaced by K⁺ in 7U9K [5–7]. For this reason, the model was generated with 4 Mg²⁺ and 2 K⁺ as co-factors and 2 ATP as substrates (D-alanine was not part of the standard list of AlphaFold ligands). The highest scoring model was selected for comparison.

The models were loaded into PyMOL, and their tertiary structures compared (Figure 2) before colouring according to either QMEANDisCo Local (homology model) (Figure 3A) or pLDDT (AI model) score (Figure 3B) to analyse per-residue local confidence and identify regions of uncertainty [8–10]. The AlphaFold model has a few areas with low confidence but only res1–6 are very low, while the Swiss-Model structure has lower confidence throughout. The AlphaFold PAE heatmap (Figure 4) highlights only 2 main areas of greater positional error at distance, these being the first few residues, previously described as having very low confidence pLDDT scores, and those around res200, with res194–206 also having low confidence pLDDT scores. The AlphaFold model's ipTM and pTM scores were 0.87 and 0.89, respectively, and although the Swiss-Model metrics of GMQE and QMEANDisCo Global (0.75 and 0.74 $\pm$ 0.05, respectively) (Table 1) aren't directly comparable, they do have the same scale and can therefore give an *approximate* comparison of overall model confidence. Taking both local and global metrics for homology and AI models, it is clear that overall confidence is higher with the AlphaFold model.

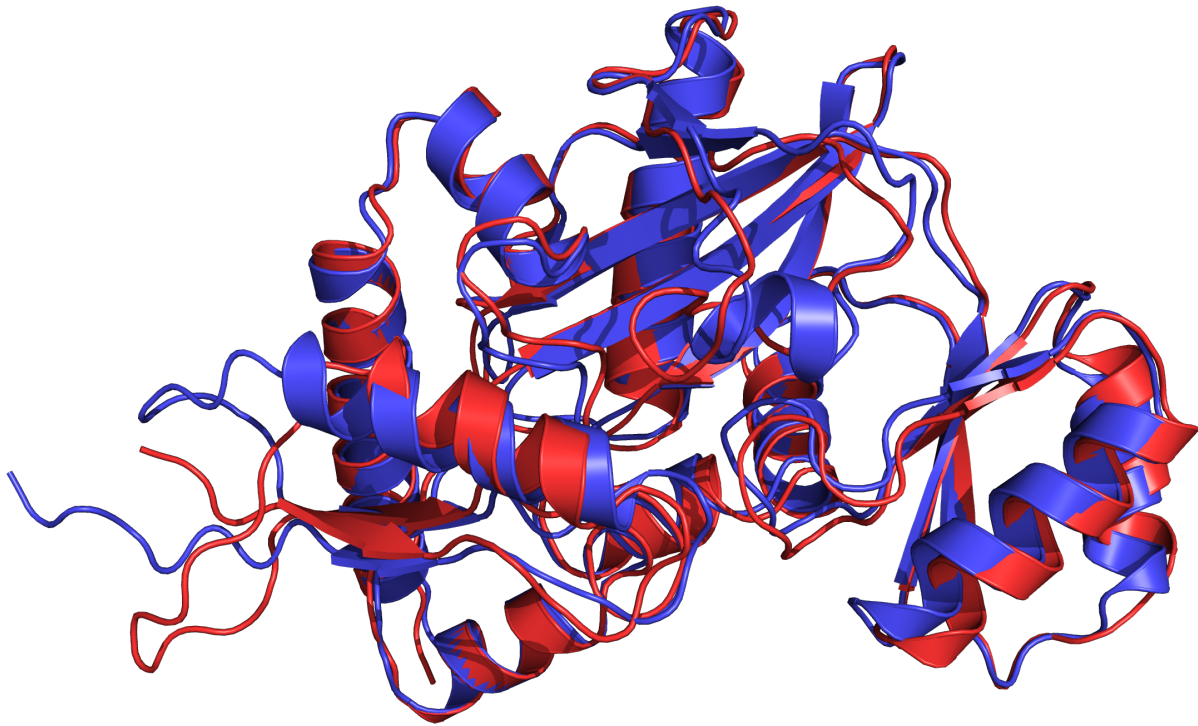


Figure 2. Tertiary structure comparison. Chain A of models produced by Swiss-Model (red) and AlphaFold3 (blue).

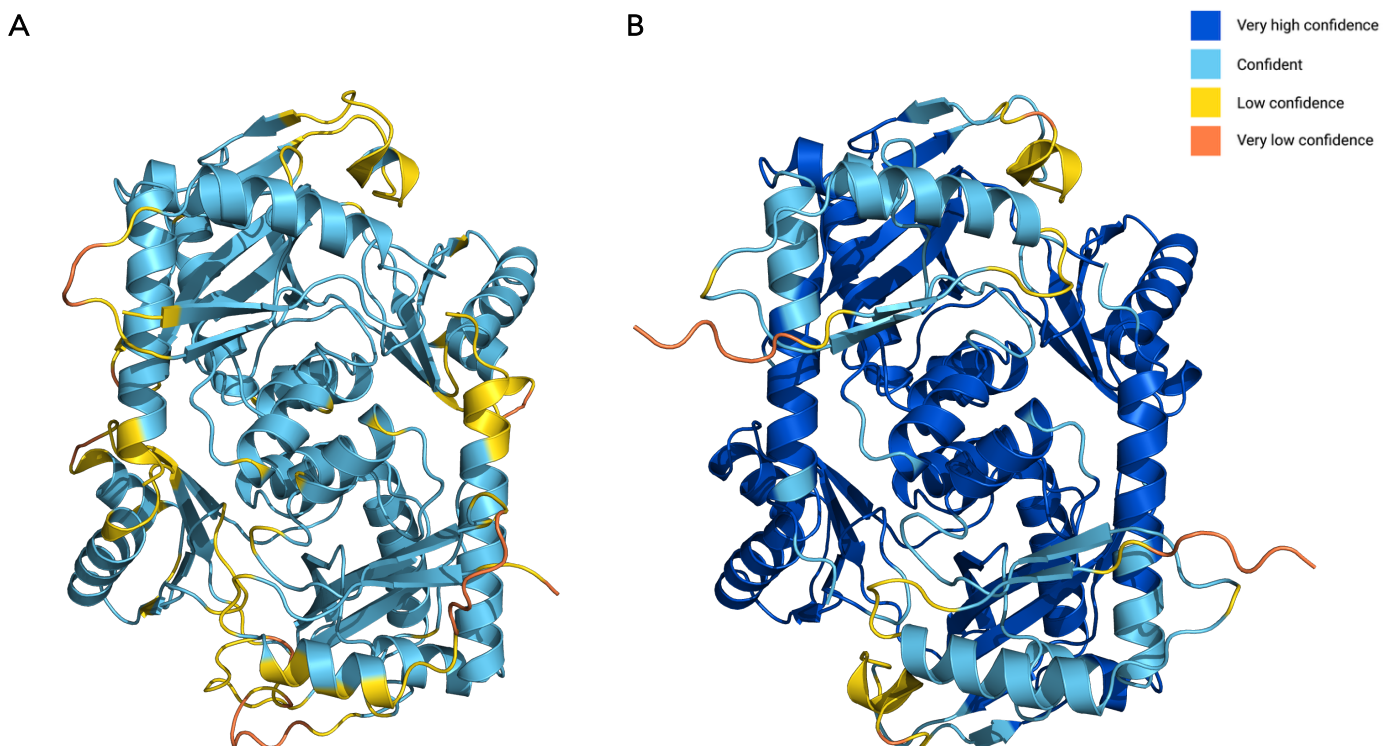


Figure 3. Predicted 3D protein structure models. Structures predicted by Swiss-Model (A) and AlphaFold (B). Local confidence is indicated by QMEANDisCo Local/pLDDT with score $> 0.9/90$ (dark blue) indicating very high confidence, $0.9/90 \geq \text{score} > 0.7/70$ (light blue) indicating confident, $0.7/70 \geq \text{score} > 0.5/50$ (yellow) indicating low confidence, and score $\leq 0.5/50$ (orange) indicating very low confidence.

2 Protein structure model comparison

Before comparing predicted models with experimentally solved structures, secondary structure elements were first visualised within the AlphaFold3 model (Figure 5). A search was then performed on the PDB website to identify experimental structures with similar sequences to that which was provided, retrieving entry 7U9K as the top result, with a percentage identity of 40% [4, 11–13]. While other entries also had similar sequence identity, 7U9K was selected due to presence of co-factors and substrates. This structure was obtained via 2.00 Å resolution X-ray diffraction, which should be sufficiently detailed to obtain an accurate structure but overall, the entry has mixed validation scores, with a relatively high Rfree score and relatively low Clashscore, while Ramachandran, Sidechain, and RSRZ outliers all fall around the average percentile ranks (Figure 6). 7U9K was coloured according to b-factor, visualising the areas of greatest experimental uncertainty (Figure 7), these primarily being found on chain B or disordered regions of protein more generally. Missing parts of the structure indicate residues that couldn't be represented because they were disordered, lacking visible electron density [4, 9].

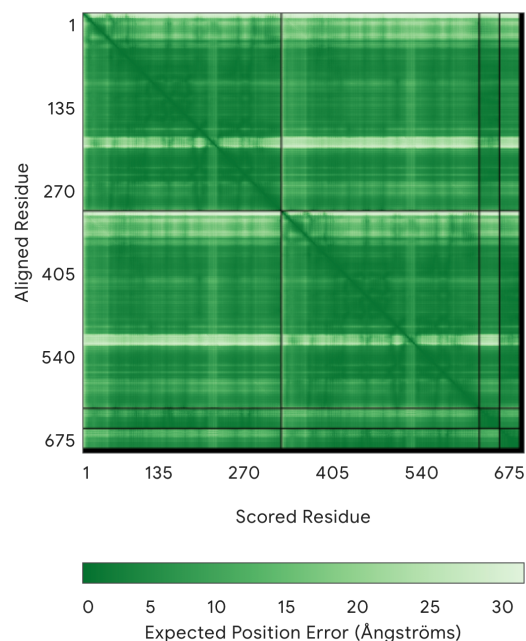


Figure 4. AlphaFold3 model PAE heatmap.

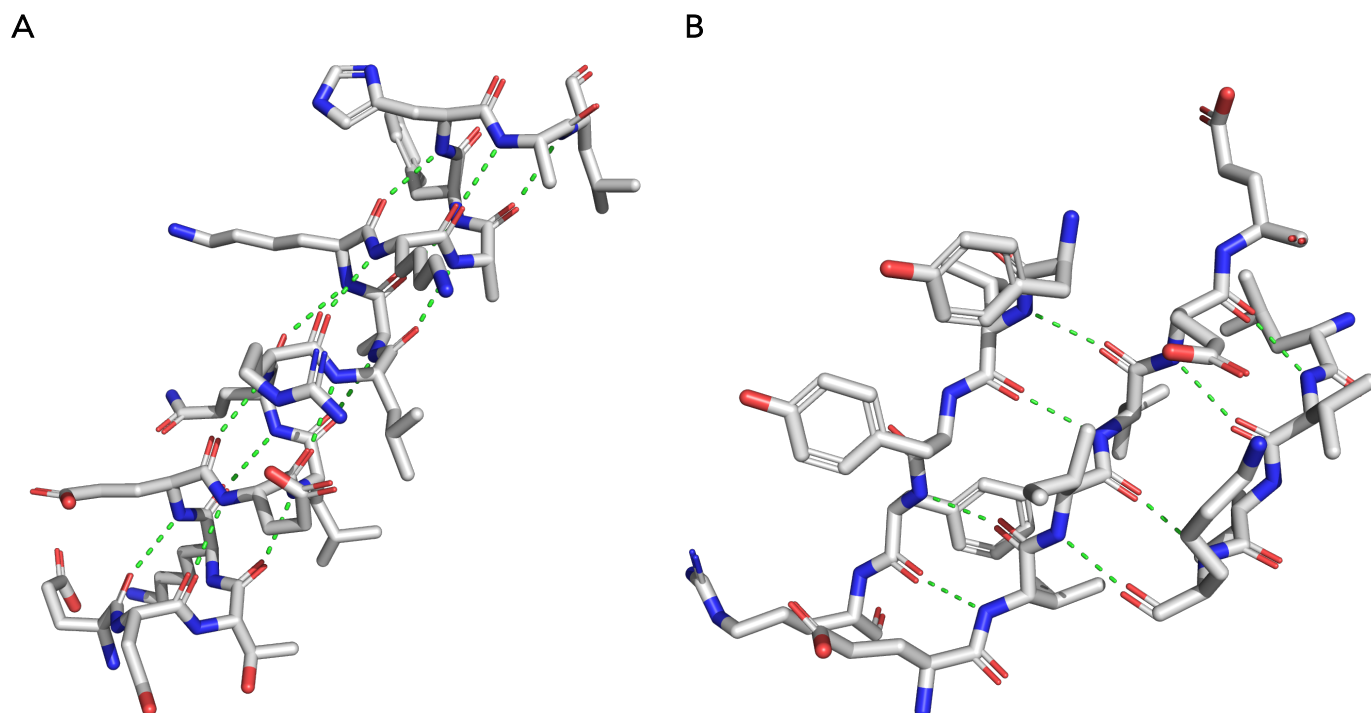


Figure 5. Secondary structure examples from AlphaFold3 model. α -helix (A) and anti-parallel β -sheet (B).

Comparing 7U9K with the predicted models, the biggest difference would be the section between residues 32 and 83, which forms a separate, external structure before then returning to a similar structure to that of the predicted models (Figure 8A). Another difference is the α -helix between 7U9K residues 151 and 171 (predicted model res103-120) that continues on for longer in 7U9K than in the predicted models. This was confirmed by closer inspection of the path taken by the protein backbone (Figure 8B). Other minor differences were present, but primarily in disordered sections of the proteins.

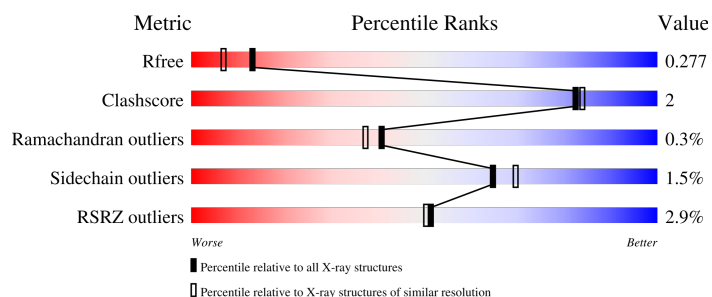


Figure 6. Validation metrics for PDB entry 7U9K.

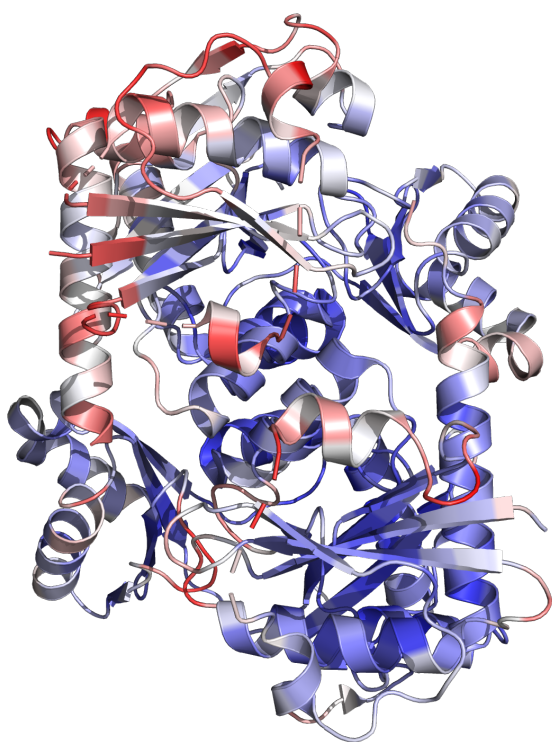


Figure 7. PDB entry 7U9K protein structure. Structure coloured according to b-factor with minimum and maximum set to 10 and 60, respectively.

3 Structure-based analysis

CATH and SCOPe classifications were identified for the folding domains present in the putative protein structure along with likely associated residues by comparing the predicted models with another homologous structure, 2I8C (Table 3) [14–20]. It is worth noting that the predicted models are missing part of the structure that would form part of the CATH 3.40.50.20 / SCOPe c.30.1.0 domain, but it is unclear whether this would significantly affect the classification. There is some consensus between CATH and SCOPe classifications in identifying the Class for each domain as mixed alpha-beta protein and the description for the Fold of c.30.1.0 also describes the presence of a Rossmann fold, similar to the Topology of 3.40.50.20. Additionally, while 3.30.470.20 and 3.30.1490.20 are split into two domains, they are both associated with ATP-grasping, similar to d.142.1.0 which is defined as an ATP-grasp Fold and covers most of the same residues.

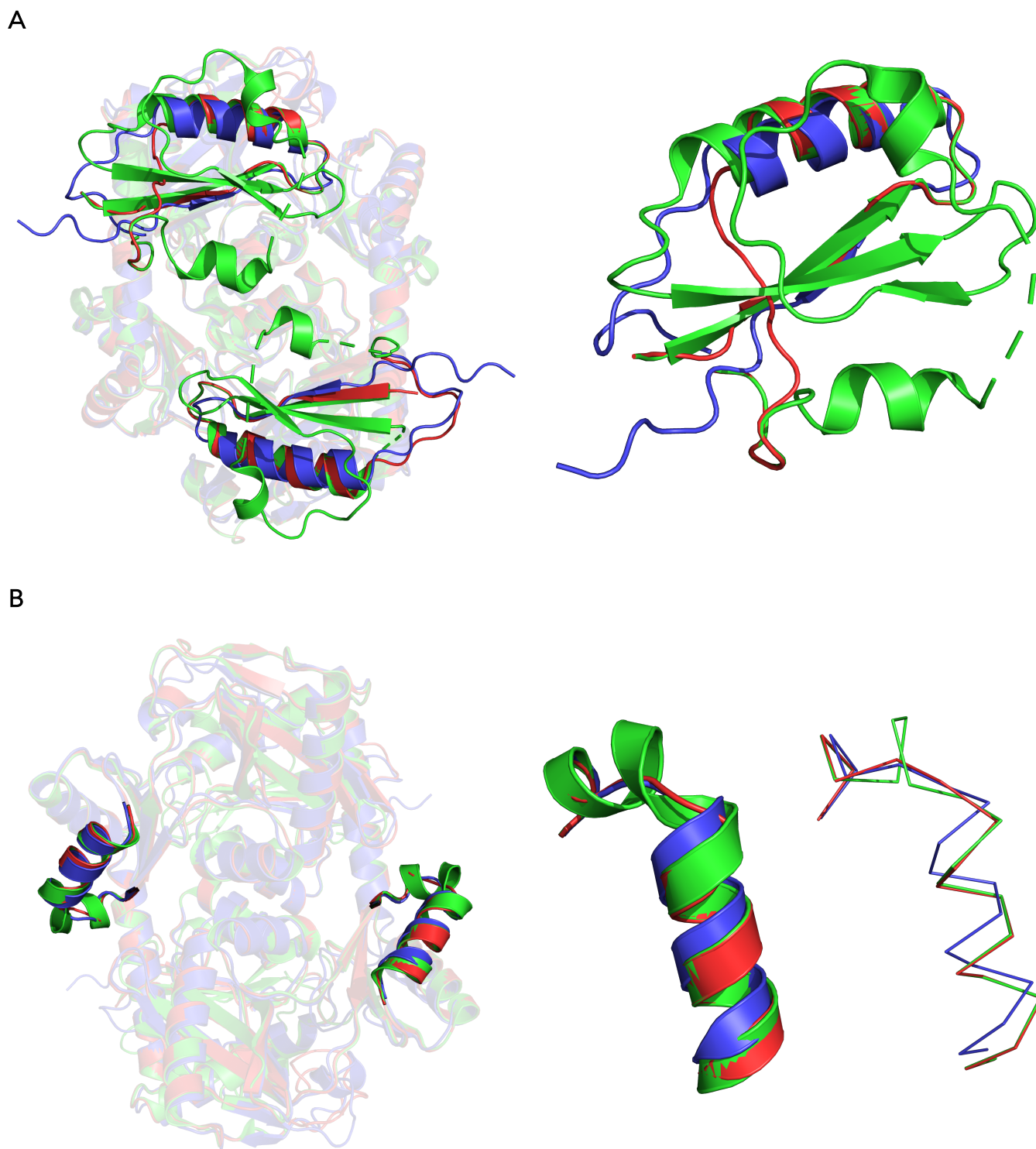


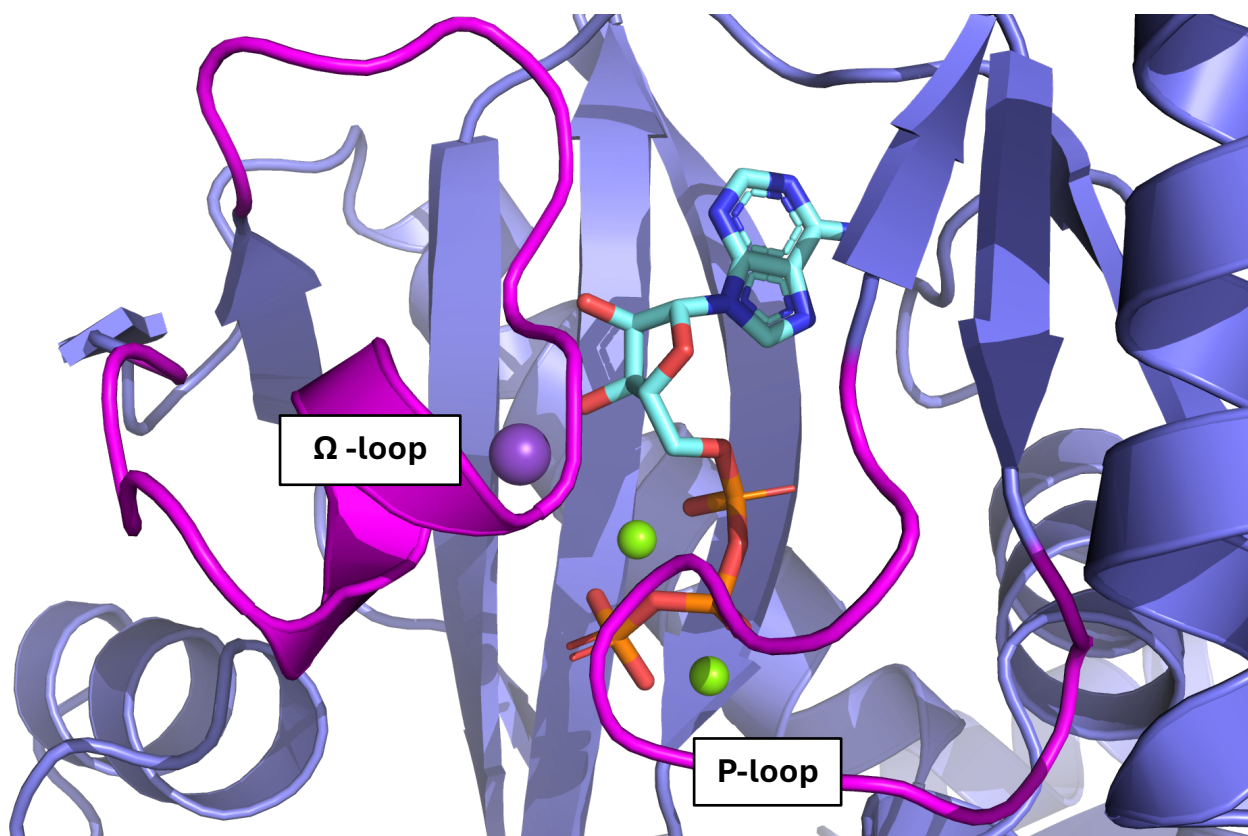
Figure 8. Differences between predicted and experimentally solved structures. Structural elements belonging to 7U9K (green) as well as Swiss-Model (red) and AlphaFold3 (blue) models. A) 7U9K res2-89 and predicted model res1-41 complete view (left) and highlighted section (chain A) (right). B) 7U9K res151-170 and predicted model res103-119 complete view (left) and highlighted section (chain A) with cartoon (centre) and ribbon (right) representations.

Table 3. CATH and SCOPe classifications.

CATH							
CATH code	Class	Architecture		Topology	Homology		Likely residues
3.40.50.20	Alpha Beta	3-Layer(aba) Sandwich		Rossmann fold			1-69
3.30.470.20	Alpha Beta	2-Layer Sandwich		D-amino Acid Aminotransferase; Chain A, domain 1	ATP-grasp fold, B domain		70-95, 167-304
3.30.1490.20	Alpha Beta	2-Layer Sandwich		Dna Ligase; domain 1	ATP-grasp fold, A domain		96-166
SCOPe							
SCOPe code	Class	Fold	Superfamily	Family	Domain	Species	Likely residues
c.30.1.0	Alpha and beta proteins (a/b)	PreATP-grasp domain	PreATP-grasp domain	Not a true family	Not a true protein	Staphylococcus aureus (TaxId: 1280)	1-80
d.142.1.0	Alpha and beta proteins (a+b)	ATP-grasp	Glutathione synthetase ATP-binding domain-like	Not a true family	Not a true protein	Staphylococcus aureus (TaxId: 1280)	81-304

The most likely function of the predicted protein is to catalyse the formation of D-ala-D-ala dipeptide, a precursor required for the biosynthesis of bacterial cell wall peptidoglycan, in an ATP-dependent 2-step reaction involving Mg^{2+} and K^+ as co-factors [6, 7, 21]. The proposed active site was explored in the AlphaFold model and key residues highlighted (Figure 9). The protein begins in an open state, with closure of the active site induced by substrate binding. ATP first binds to one pocket of the active site, facilitated by Mg^{2+} cations acting as catalytic binding anchors for the triphosphate group, triggering the closure of the first flexible loop region (P-loop) over the triphosphate group. Hydrogen bonding between ATP and residues E162, I165, and E169 stabilises the adenine group. The triphosphate group is stabilised by hydrogen bonding and/or ionic interactions with K126, S132, G133, Y202, and N258. The first D-alanine then docks within another pocket, triggering closure of the second loop (Ω -loop). D-alanine is stabilised by hydrogen bonds with Y202 and R241 and hydrophobic interaction with F48. The reaction can then occur, with the D-alanine first being phosphorylated by ATP, before the phosphorylated intermediate then reacts with a second D-alanine (guided by hydrogen bonding with primarily R241), releasing the phosphate and producing the D-ala-D-ala dipeptide [6]. The K^+ cation, although not strictly required for the reaction to occur, does increase the rate at which it proceeds [6, 21].

A



B

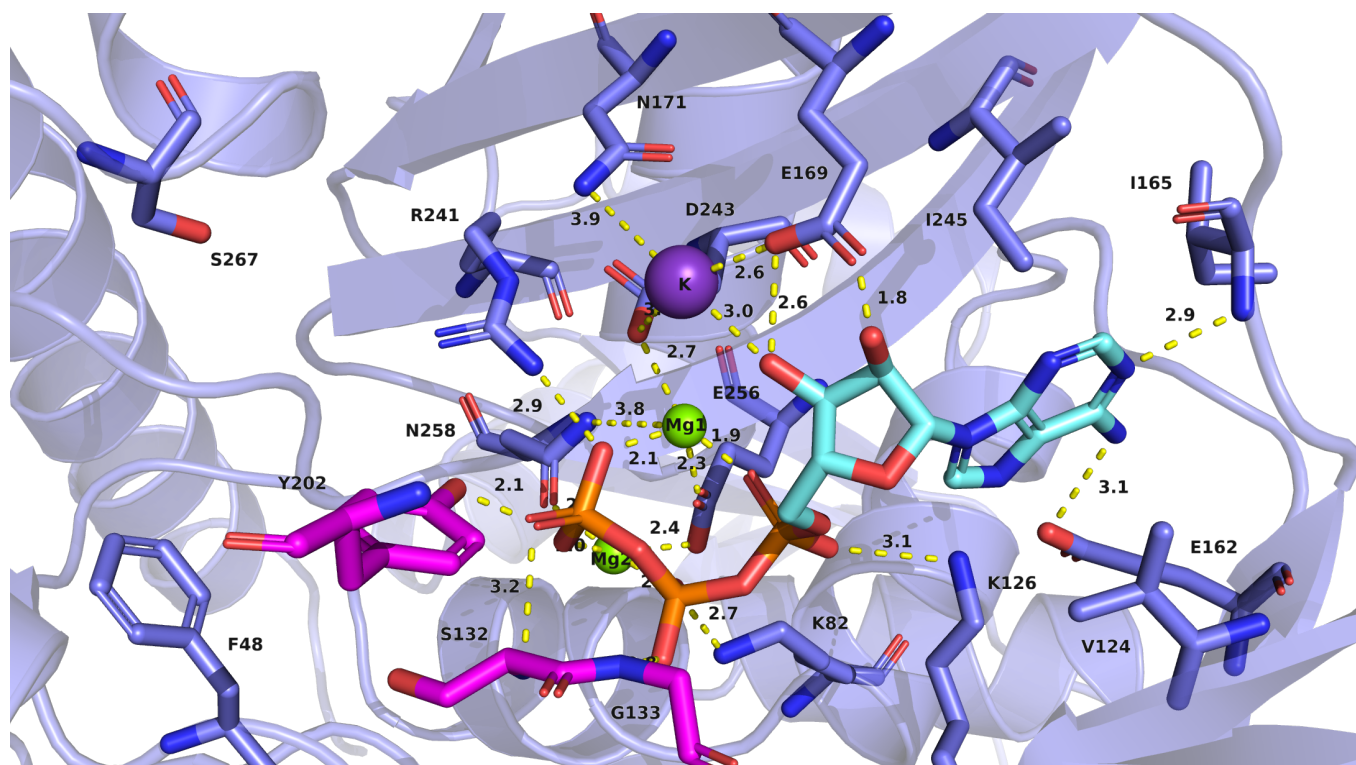


Figure 9. Active site of AlphaFold3 model. AlphaFold3 model (slate) with P-loop and Ω -loop highlighted (magenta), ATP (cyan), and spherically represented cations Mg^{2+} (green) and K^{+} (purple). A) Complete view of active site with loops closed. B) Closer view of active site with key residues labelled and represented as sticks as well as key interactions indicated. Loop cartoons have been excluded here for clarity.

4 Structure-based predictions

The protein's main global motions were visualised using the iMOD server, covering the first 3 modes that together account for over 40% of the variance (Figure 10E) [22]. Each of these modes can be divided into two clusters of movement that broadly align with the movement of each chain. The first mode describes a twisting motion where the clusters rotate in opposite directions (Figure 10A). The second and third describe see-saw motions where either end of the cluster rock back and forth towards each other (Figure 10B and 10C). It is proposed that these combined motions allow the protein to capture substrates within the active site and close around them, making them react.

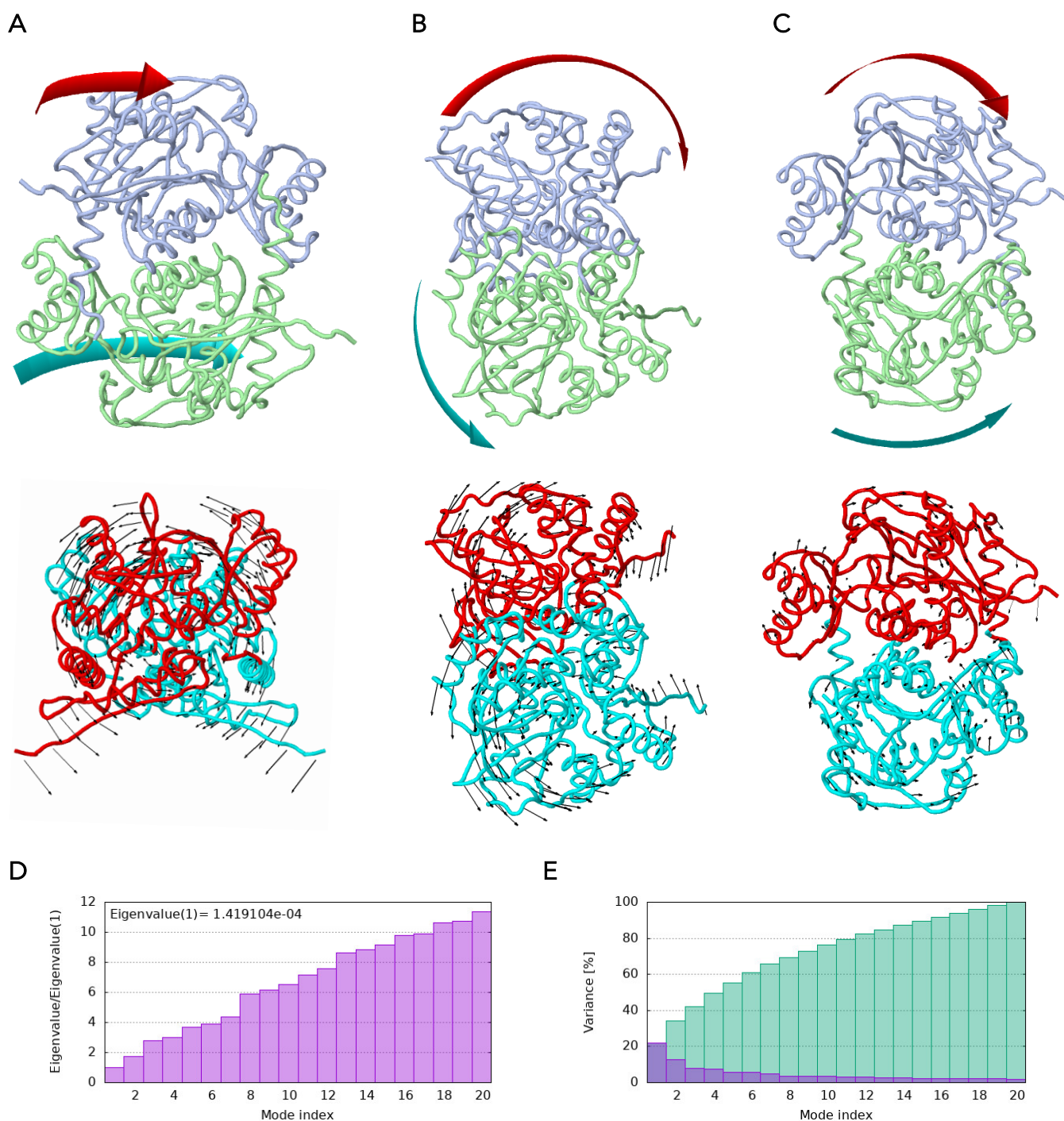


Figure 10. Global protein motions of the AlphaFold3 model. Global motion modes 1 (A), 2 (B), and 3 (C) visualised by chain with affine arrows (top) and by cluster with arrow field (bottom). Eigenvalues (E) and % Variance (F) included.

Optimal ligand binding poses within the AlphaFold model active site (with ATP removed) were predicted using SwissDock to perform a Molecular Docking computation and top scoring poses were compared [23, 24]. The highest scoring poses, according to AC and SwissParam scores, were both positioned in similar poses where ATP should bind within the active site, the only significant difference being a slight rotation of the phosphate group. The optimal ligand binding pose is suggested as follows (Figure 11), this being the pose that had the highest AC score. Key to ligand binding in this pose are hydrogen bonds and ionic interactions with S132, Y202, R241, and the 2 Mg^{2+} cations. The potential effects of single amino acid mutations on ligand binding were also examined. Exchanging E256 or N258 with an alanine would greatly reduce the stability of both Mg^{2+} cations, consequently impacting that of the ligand. Replacing Y202 with phenylalanine would stop it forming hydrogen bonds with the ligand, so the ring might rotate, as either end is pulled towards S132 and R241. This might then lead to additional π - π interactions with phenylalanine.

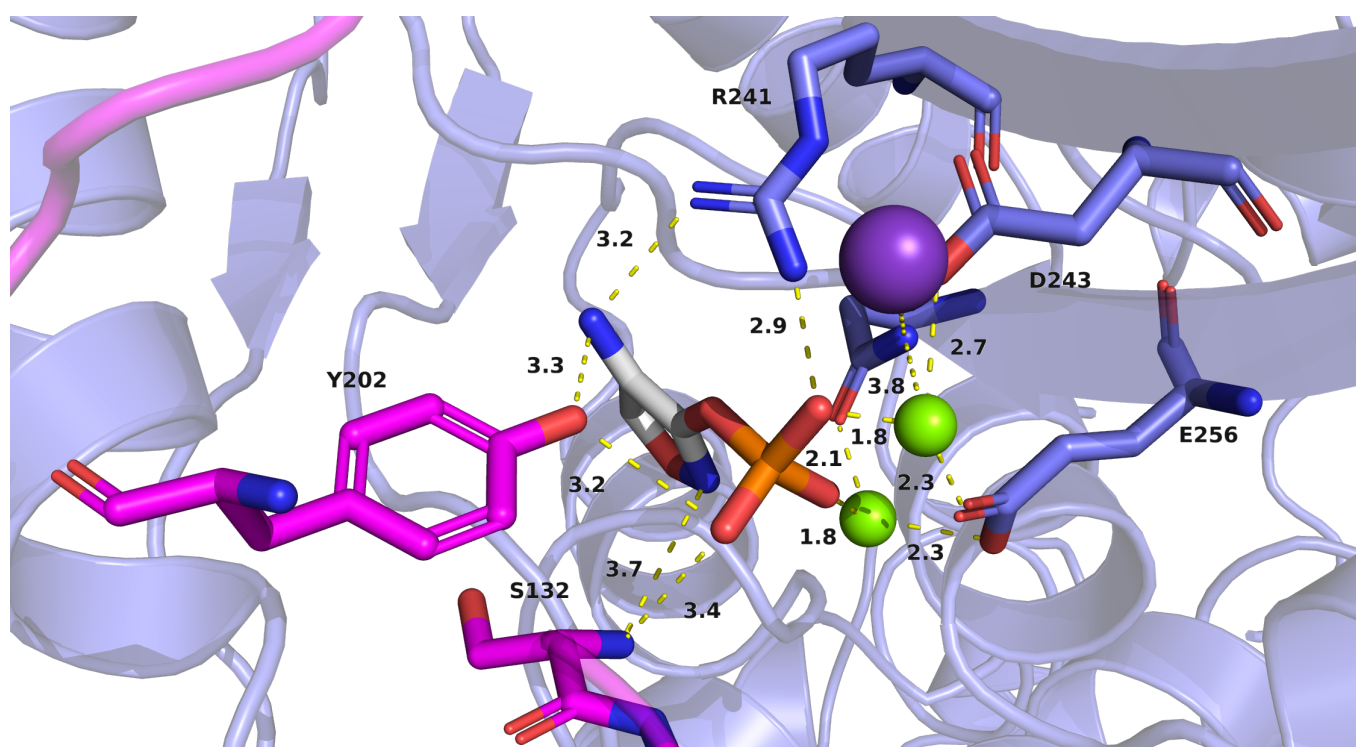


Figure 11. Optimal ligand binding pose in AlphaFold3 model. Ligand (white) in in AlphaFold3 model (slate) active site with loop residues highlighted (magenta) and spherical representations for Mg^{2+} (green) and K^{+} (purple) cations.

5 Conclusion

Of the predicted models generated, the AlphaFold3 model was found to have the greatest overall accuracy. It was also determined that the most similar experimental structure, 7U9K, had some structural differences, with an entire section missing from the predicted models. Examining the CATH and SCOPe classifications of 2I8C helped to elucidate the functional domains of the predicted protein, revealing that it is most likely a DDI enzyme, catalysing the production of D-ala-D-ala dipeptide. Examining the active site demonstrated the importance of the 2 Mg^{2+}

cations and key residues in stabilising ATP and D-alanine within their respective pockets. Analysis of the global movements of the protein were likely required to capture and react these substrates within the active site. Molecular Docking analysis with the provided ligand shows that it is most likely an inhibitor of the enzyme, with the top pose binding in place of ATP. Mutation of residues associated with binding could affect ligand stability or rotation within the active site.

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